Product Specifications

Lecture 4

Outline

- Problem Definition
- Determining Product Specifications
- Quality Function Deployment Matrix
- Classroom Activity: Determine Specs of Backpack
- Engineering Idea #3: Kidney Exchange

Problem Definition

- Formal Description of the task (mission) to be solved
- Also referred to as ‘problem statement’, ‘mission statement’
- Formal
  - In written form
  - Approved by appropriate people (client, manager, etc.)
- Starting point of overall engineering design process

Problem Definition

- Problem = task to be solved
- Do not describe difficulties or weakness as problem definition

  ➔ Describe what to do for difficulties or weakness

  eg) Current railway transportation is very slow (X)

  Improve the speed of railway transportation (O)
- Do not use subjective and relative terms in problem definition

  ➔ Describe objective and verifiable terms

  eg) Build an archiving system with large storage (X)

  Build an archiving system with storage of at least 5TB (O)

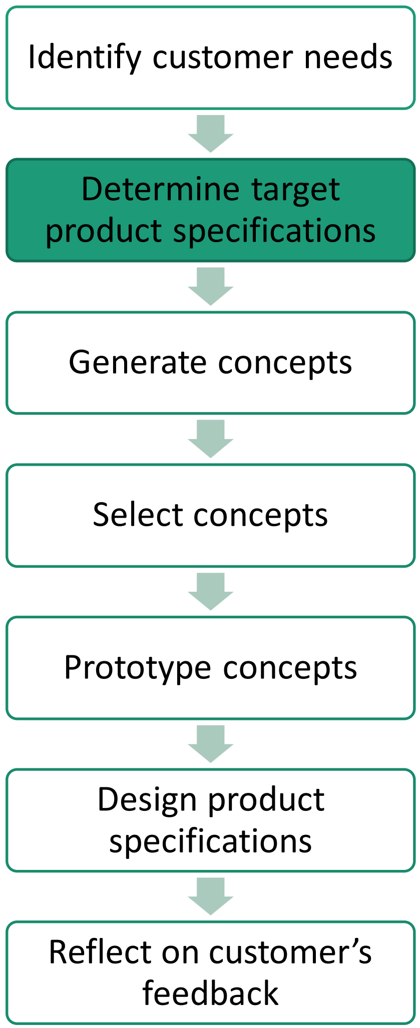
Problem Definition

- Purpose of problem definition
  - To clarify the purpose of design among many stakeholders (clients, users, managers, developers, etc.)
  - To provide guidelines for engineering design work
  - To verify the success of final design products
- Problem definition can be hierarchical
  - Problem Definition for the whole project or whole teams
  - Problem Definition for sub-projects or particular team

Determining Product Specifications

Fork suspension

Determining Product Specifications



- Translating relatively subjective customer's needs into precise targets
- Determining the factors which constitute the success or failure of product design
- Find dominant design parameters and constraints
- Maximizing the impact the product makes through minimizing the effort required for development
- Resolving trade-offs between various product characteristics.

Steps for Determining Product Specifications

1. Identify a list of metrics and measurement units that sufficiently address the needs
2. Link metrics and measurements to needs
3. Collect the competitive benchmarking information
4. Assign ideal and marginally acceptable target values for each metric (using at least, at most, between and exactly, etc)

Product Specifications Example: Mountain Bike Suspension Fork

Start with the Customer Needs

#	NEED		Imp
1	The suspension	reduces vibration to the hands.	3
2	The suspension	allows easy traversal of slow, difficult terrain.	2
3	The suspension	enables high speed descents on bumpy trails.	5
4	The suspension	allows sensitivity adjustment.	3
5	The suspension	preserves the steering characteristics of the bike.	4
6	The suspension	remains rigid during hard cornering.	4
7	The suspension	is lightweight.	4
8	The suspension	provides stiff mounting points for the brakes.	2
9	The suspension	fits a wide variety of bikes, wheels, and tires.	5
10	The suspension	is easy to install.	1
11	The suspension	works with fenders.	1
12	The suspension	instills pride.	5
13	The suspension	is affordable for an amateur enthusiast.	5
14	The suspension	is not contaminated by water.	5
15	The suspension	is not contaminated by grunge.	5
16	The suspension	can be easily accessed for maintenance.	3
17	The suspension	allows easy replacement of worn parts.	1
18	The suspension	can be maintained with readily available tools.	3
19	The suspension	lasts a long time.	5
20	The suspension	is safe in a crash.	5

Metrics Exercise

Customer need for a ballpoint pen:

- The pen writes smoothly

Metrics Exercise

Customer need for a ballpoint pen:

- The pen writes smoothly

Metrics (or design variables):

- Variation in line thickness - mm
- Variation in ink coverage - cc/mm2
- Functional range of writing force - N
- Functional range of writing velocity - mm/sec
- Functional range of pen angle from vertical - deg
- Variation in resistance to translational motion - N

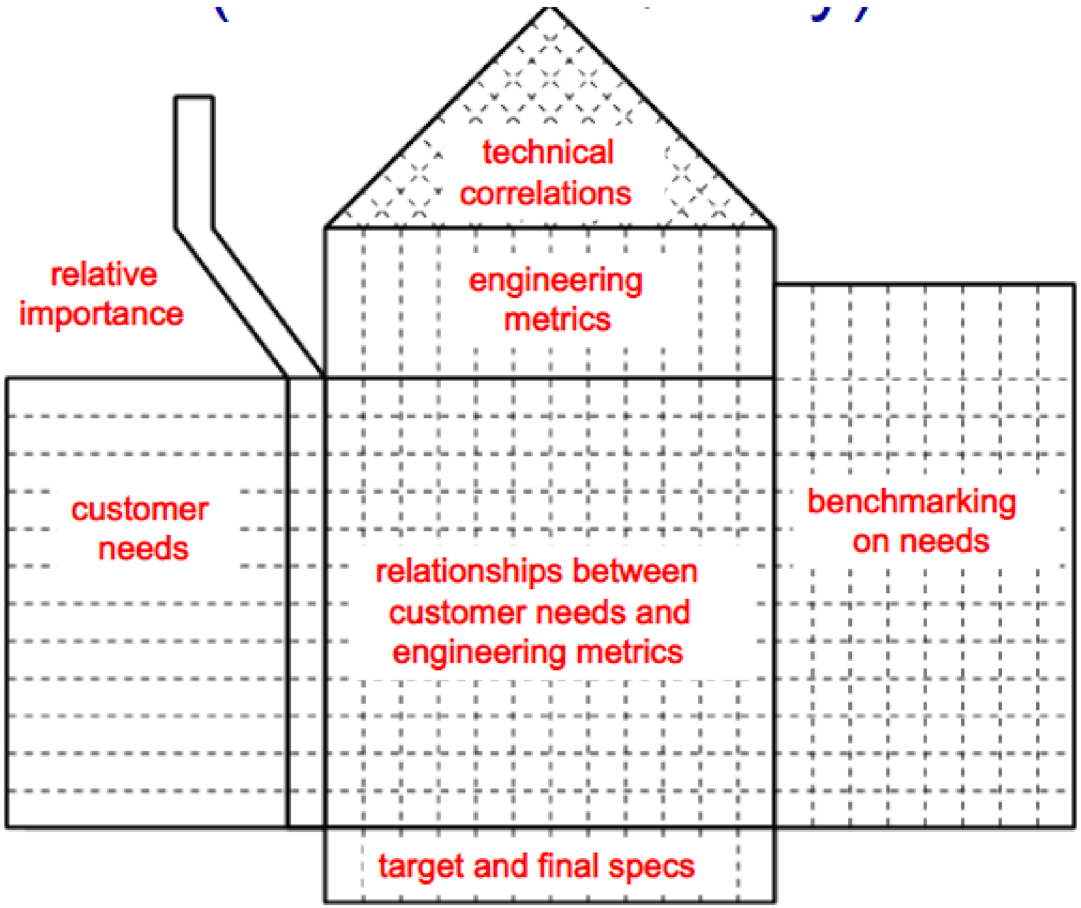
1. Establish Metric and Units

Metric #	Need #s	Metric	Imp	Units
1	1,3	Attenuation from dropout to handlebar at 10hz	3	dB
2	2,6	Spring pre-load	3	N
3	1,3	Maximum value from the Monster	5	g
4	1,3	Minimum descent time on test track	5	s
5	4	Damping coefficient adjustment range	3	N-s/m
6	5	Maximum travel (26in wheel)	3	mm
7	5	Rake offset	3	mm
8	6	Lateral stiffness at the tip	3	kN/m
9	7	Total mass	4	kg
10	8	Lateral stiffness at brake pivots	2	kN/m
11	9	Headset sizes	5	in
12	9	Steertube length	5	mm
13	9	Wheel sizes	5	list
14	9	Maximum tire width	5	in
15	10	Time to assemble to frame	1	s
16	11	Fender compatibility	1	list
17	12	Instills pride	5	subj
18	13	Unit manufacturing cost	5	US$
19	14	Time in spray chamber w/o water entry	5	s
20	15	Cycles in mud chamber w/o contamination	5	k-cycles
21	16,17	Time to disassemble/assemble for maintenance	3	s
22	17,18	Special tools required for maintenance	3	list
23	19	UV test duration to degrade rubber parts	5	hours
24	19	Monster cycles to failure	5	cycles
25	20	Japan Industrial Standards test	5	binary
26	20	Bending strength (frontal loading)	5	MN

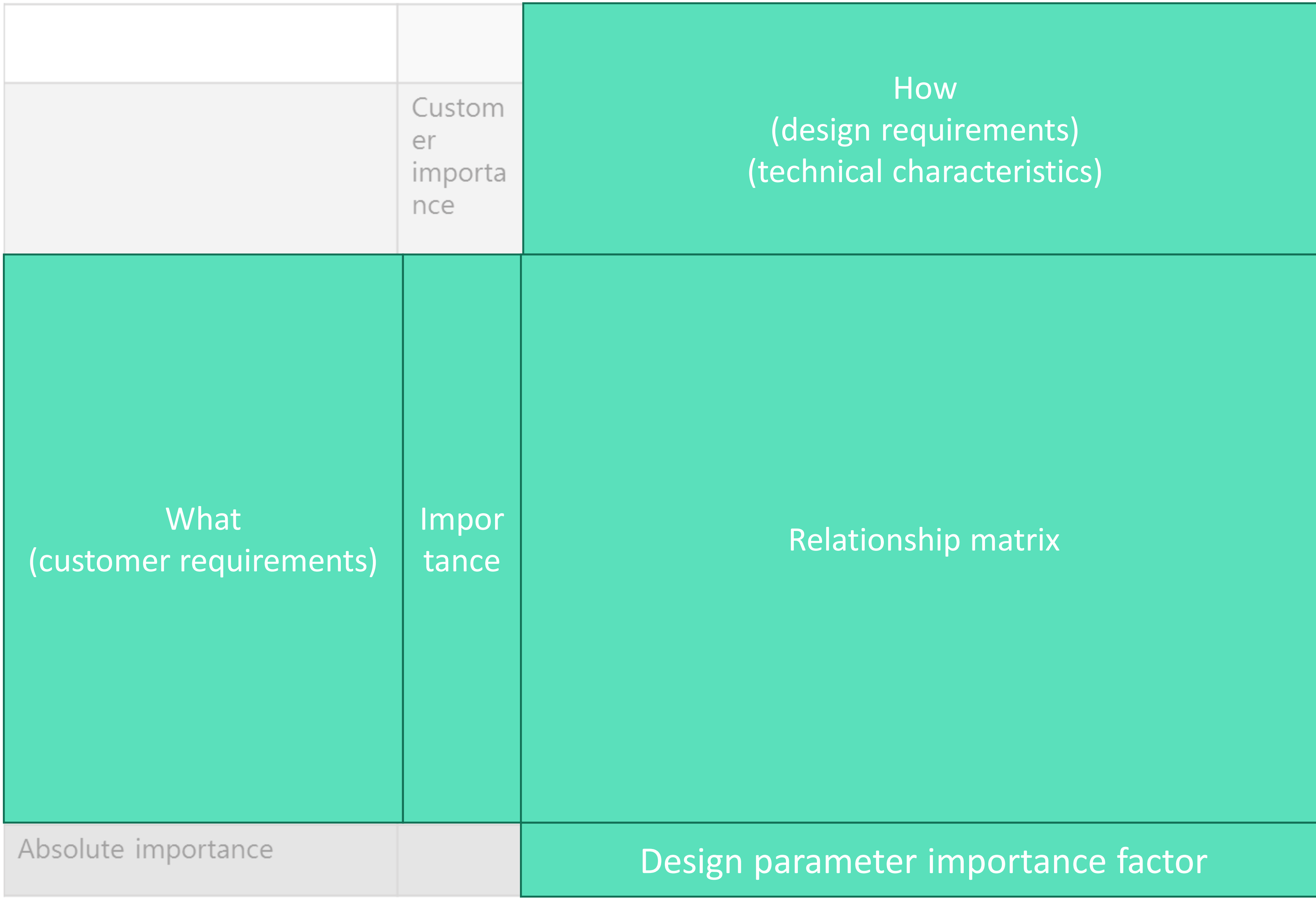
2. Link Metrics to Needs

Need		1 Attenuation from dropout to handlebar at 10hz	2 Spring pre-load	3 Maximum value from the Monster	4 Minimum descent time on test track	5 Damping coefficient adjustment range	6 Maximum travel (26in wheel)	7 Rake offset	8 Lateral stiffness at the tip	9 Total mass	10 Lateral stiffness at brake pivots	11 Headset sizes	12 Steertube length	13 Wheel sizes	14 Maximum tire width	15 Time to assemble to frame	16 Fender compatibility	17 Instills pride	18 Unit manufacturing cost	19 Time in spray chamber w/o water entry	20 Cycles in mud chamber w/o contamination	21 Time to disassemble/assemble for maintenance	22 Special tools required for maintenance	23 UV test duration to degrade rubber parts	24 Monster cycles to failure	25 Japan Industrial Standards test	26 Bending strength (frontal loading)
1	reduces vibration to the hands.	•		•	•																						
2	allows easy traversal of slow, difficult terrain.		•																								
3	enables high speed descents on bumpy trails.	•		•	•																						
4	allows sensitivity adjustment.					•																					
5	preserves the steering characteristics of the bike.						•	•																			
6	remains rigid during hard cornering.		•						•																		
7	is lightweight.									•																	
8	provides stiff mounting points for the brakes.										•																
9	fits a wide variety of bikes, wheels, and tires.											•	•	•	•												
10	is easy to install.															•											
11	works with fenders.																•										
12	instills pride.																	•									
13	is affordable for an amateur enthusiast.																		•								
14	is not contaminated by water.																			•							
15	is not contaminated by grunge.																				•						
16	can be easily accessed for maintenance.																					•					
17	allows easy replacement of worn parts.																					•	•				
18	can be maintained with readily available tools.																						•				
19	lasts a long time.																							•	•		
20	is safe in a crash.																									•	•

Quality Function Deployment (House of Quality)



QFD structure Without Benchmarking



QFD Example: Climbing harness

Strong(9) ◉ Medium(4) ○ Weak(1) △

Quality function deployment

		Performance measure			Size of range		Technical details		
	Customer importance	Meet standard	Harness weight	Webbing strength	No. of colors	No. of sizes	Padding thickness	No. of buckles	No. of gear loops
Easy to put on	2					○		◉	
Comfortable when hanging	5					○	◉	○	
Fits over different clothes	1					○	○	◉	
Accessible gear loops	3								◉
Does not restrict movement	5		○			○	◉	○	
Light weight	3		◉	○			○	△	△
Safe	5	◉	○	◉					
Attractive	2		△		◉		△	△	
Absolute importance		45	69	57	18	52	108	72	30

Quality function deployment – step 1

		Performance measure			Size of range		Technical details		
	Customer importance	Meet standard	Harness weight	Webbing strength	No. of colors	No. of sizes	Padding thickness	No. of buckles	No. of gear loops
Easy to put on	2					○		◎	
Comfortable when hanging	5					○	◎	○	
Fits over different clothes	1					○	○	◎	
Accessible gear loops	3								◎
Does not restrict movement	5		○			○	◎	○	
Light weight	3		◎	○			○	△	△
Safe	5	◎	○	◎					
Attractive	2		△		◎		△	△	
Absolute importance		45	67	57	18	52	108	72	30

	Customer importance
Easy to put on	2
Comfortable when hanging	5
Fits over different clothes	1
Accessible gear loops	3
Does not restrict movement	5
Light weight	3
Safe	5
Attractive	2

- Create a list of customer requirements (WHATs)
- Record customer responses to the question: “What are the important (qualities, characteristics, elements, features) of XXX ?
- Record in customer’s own words (“Voice of the Customer”)
- Internal & External customers
- Prioritize the customer requirements on a scale of 1-5

Quality function deployment – step 2

		Performance measure			Size of range		Technical details		
	Customer importance	Meet standard	Harness weight	Webbing strength	No. of colors	No. of sizes	Padding thickness	No. of buckles	No. of gear loops
Easy to put on	2					○		◎	
Comfortable when hanging	5					○	◎	○	
Fits over different clothes	1					○	○	◎	
Accessible gear loops	3								◎
Does not restrict movement	5		○			○	◎	○	
Light weight	3		◎	○			○	△	△
Safe	5	◎	○	◎					
Attractive	2		△		◎		△	△	
Absolute importance		45	67	57	18	52	108	72	30

- Compile a list of the design requirements and technical characteristics (HOWs) necessary to achieve the customer's requirements

Performance measure			Size of range		Technical details		
Meet standard	Harness weight	Webbing strength	No. of colors	No. of sizes	Padding thickness	No. of buckles	No. of gear loops

Quality function deployment – step 2

		Performance measure			Size of range		Technical details		
	Customer importance	Meet standard	Harness weight	Webbing strength	No. of colors	No. of sizes	Padding thickness	No. of buckles	No. of gear loops
Easy to put on	2					○		◉	
Comfortable when hanging	5					○	◉	○	
Fits over different clothes	1					○	○	◉	
Accessible gear loops	3								◉
Does not restrict movement	5		○			○	◉	○	
Light weight	3		◉	○			○	△	△
Safe	5	◉	○	◉					
Attractive	2		△		◉		△	△	
Absolute importance		45	67	57	18	52	108	72	30

- Function/Performance
- Product Cost
- Lead time (Time-to-market)
- Quantity
- Environmental effect
- Safety
- Quality
- Energy efficiency
- Reliability
- Maintenance
- Mechanical weight
- Size
- Space constraints
- Aesthetics
- Packaging and delivery
- Human resource
- Product life time
- Operation noise
- Operating guide
- Human factors
- Health
- Government regulations
- Operation cost

Performance measure			Size of range		Technical details		
Meet standard	Harness weight	Webbing strength	No. of colors	No. of sizes	Padding thickness	No. of buckles	No. of gear loops

Quality function deployment – step 3

		Performance measure			Size of range		Technical details		
	Customer importance	Meet standard	Harness weight	Webbing strength	No. of colors	No. of sizes	Padding thickness	No. of buckles	No. of gear loops
Easy to put on	2					○		◉	
Comfortable when hanging	5					○	◉	○	
Fits over different clothes	1					○	○	◉	
Accessible gear loops	3								◉
Does not restrict movement	5		○			○	◉	○	
Light weight	3		◉	○			○	△	△
Safe	5	◉	○	◉					
Attractive	2		△		◉		△	△	
Absolute importance		45	67	57	18	52	108	72	30

- Determine relationship of design requirements to customer requirements

		Performance measure			Size of range		Technical details		
	Customer importance	Meet standard	Harness weight	Webbing strength	No. of colors	No. of sizes	Padding thickness	No. of buckles	No. of gear loops
Easy to put on	2					○		◉	
Comfortable when hanging	5					○	◉	○	
Fits over different clothes	1					○	○	◉	
Accessible gear loops	3								◉
Does not restrict movement	5		○			○	◉	○	
Light weight	3		◉	○			○	△	△
Safe	5	◉	○	◉					
Attractive	2		△		◉		△	△	

Strong (9) ◉

Medium (4) ○

Weak (1) △

Quality function deployment – step 4

- Evaluate design parameter importance factor → Key design parameters

		Performance measure			Size of range		Technical details		
	Customer importance	Meet standard	Harness weight	Webbing strength	No. of colors	No. of sizes	Padding thickness	No. of buckles	No. of gear loops
Easy to put on	2					○		◎	
Comfortable when hanging	5					○	◎	○	
Fits over different clothes	1					○	○	◎	
Accessible gear loops	3								◎
Does not restrict movement	5		○			○	◎	○	
Light weight	3		◎	○			○	△	△
Safe	5	◎	○	◎					
Attractive	2		△		◎		△	△	
Absolute importance		45	67	57	18	52	108	72	30

		Performance measure			Size of range		Technical details		
	Customer importance	Meet standard	Harness weight	Webbing strength	No. of colors	No. of sizes	Padding thickness	No. of buckles	No. of gear loops
Easy to put on	2					○		◎	
Comfortable when hanging	5					○	◎	○	
Fits over different clothes	1					○	○	◎	
Accessible gear loops	3								◎
Does not restrict movement	5		○			○	◎	○	
Light weight	3		◎	○			○	△	△
Safe	5	◎	○	◎					
Attractive	2		△		◎		△	△	
Absolute importance		45	67	57	18	52	108	72	30

Quality function deployment

Strong(9) ◉ Medium(4) ○ Weak(1) △

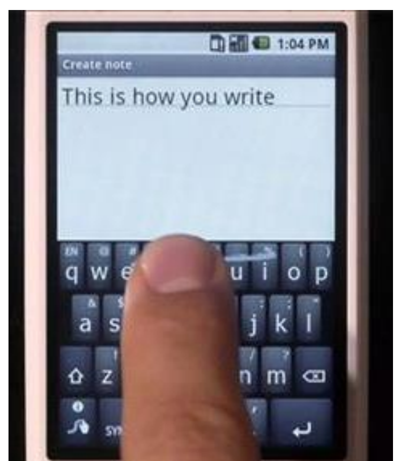
		Performance measure			Size of range		Technical details		
	Customer importance	Meet standard	**Harness weight**	Webbing strength	No. of colors	No. of sizes	**Padding thickness**	**No. of buckles**	No. of gear loops
Easy to put on	2					○		◉	
Comfortable when hanging	5					○	◉	○	
Fits over different clothes	1					○	○	◉	
Accessible gear loops	3								◉
Does not restrict movement	5		○			○	◉	○	
Light weight	3		◉	○			○	△	△
Safe	5	◉	○	◉					
Attractive	2		△		◉		△	△	
Absolute importance		45	**69**	57	18	52	**108**	**72**	30

Quality function deployment: (Handheld electric device)

Strong (9)	Medium (4)	Weak (1)
◉	○	△

		User experience			Electric Details			
		Number of buttons	Shape	Weight	Power consumptio n	Temperature characteristics	Operating voltage range	Operating frequency
Easy to carry	Comfortable to carry		◉	○				
	Convenient as it is small			△				
	Convenient as it is light			◉				
	Safe to handle		◉					
Easy to use	Proper weight			◉				
	Proper size			△				
	Comprehensive to use	◉						

QFD Example: Keyboard on smartphone



Quality function deployment: Keyboard

First step: Collect customer needs
(What are the problems of conventional keyboards?)

- Typo rate increases as the mechanical structure degrades over time
- Key assignments are fixed and cannot be modified
- Occupies a large area especially for laptops
- In smart phones, key pads are too small
- Too heavy
- Printed letters on the key pad wear out over time
- Public keyboards are not sanitary
- Dirt collects in the space between the key pads
- Too expensive

Quality function deployment: Keyboard

Second step: Transform into engineering terms

- Permanent use (C1)
- Flexible key assignment (C2)
- Reduce keyboard area for laptop (C3)
- Flexible size of key pad (C4)
- Permanent printed letter on key pad (C5)
- Maintain sanitation even for public keyboard (C6)
- Remove the space between the key pads to prevent dirt (C7)
- Reduce cost (C8)

Quality function deployment: Keyboard

Third step: Define design specifications

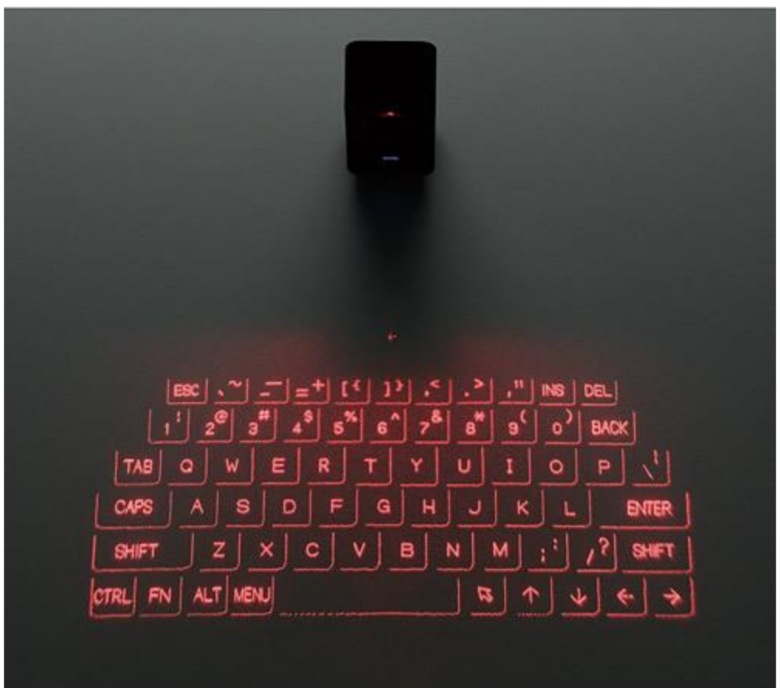
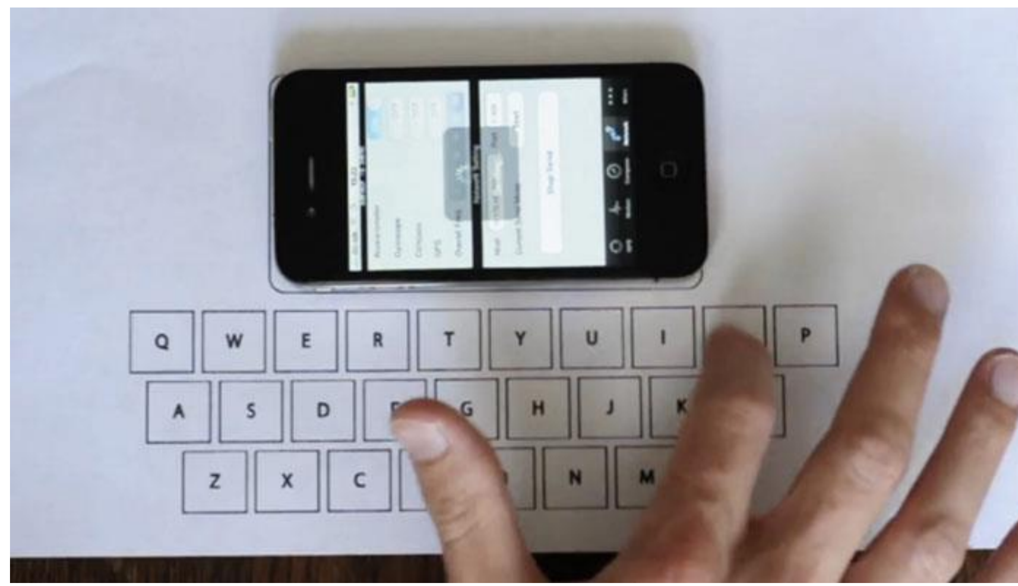
- Include technical and non-technical requirements and constraints
  - Technical: Power consumption, Maintenance, …
  - Non-technical: Cost, Aesthetics, …
- For Keyboard:
  - Convenience, volume, weight, compatibility, cost, reliability, maintenance, production cost, safety, power consumption, aesthetics, noise, …

Quality function deployment: Keyboard

Fourth step: QFD

	Convenience	Volume	Weight	Compatibility	Cost	Reliability	Maintenance	Production time	Safety	Power consumption	Aesthetics	Noise	Total
C1	0	0	0	0	2	5	4	0	0	0	0	0	**11**
C2	5	3	3	3	0	2	3	3	0	0	3	0	**25**
C3	5	5	5	1	3	3	1	0	0	3	3	0	**29**
C4	5	3	3	5	2	1	0	0	0	0	0	0	**19**
C5	5	3	3	5	0	0	0	0	0	0	3	0	**19**
C6	0	0	0	0	0	5	3	0	0	2	3	3	**16**
C7	5	5	5	0	3	3	0	0	0	0	3	0	**24**
C8	3	0	0	0	0	5	3	0	0	0	3	1	**15**
C9	5	3	3	5	5	2	2	0	0	0	2	0	**27**
Total	**33**	**22**	**22**	**19**	**15**	**26**	**16**	**3**	**0**	**5**	**20**	**4**	


one2TOUCH
Q W E R
A S D F G
Z X C V B


ESC INS DEL
BACK
TAB Q W E R T Y U I O P
CAPS A S D F G H J K L ENTER
SHIFT Z X C V B N M SHIFT
CTRL FN ALT MENU

Q W E R T Y U I P
A S D G H J K
Z X C N M

Importance of QFD

		Performance measure			Size of range		Technical details		
	Customer importance	Meet standard	Harness weight	Webbing strength	No. of colors	No. of sizes	Padding thickness	No. of buckles	No. of gear loops
Easy to put on	2					○		◉	
Comfortable when hanging	5					○	◉	○	
Fits over different clothes	1					○	○	◉	
Accessible gear loops	3								◉
Does not restrict movement	5		○			○	◉	○	
Light weight	3		◉	○			○	△	△
Safe	5	◉	○	◉					
Attractive	2		△		◉		△	△	
Absolute importance		45	67	57	18	52	108	72	30

- Provides a framework for upfront planning and product development
- Use multi-functional teams to enhance design and decision-making
- Promotes teamwork (necessary for concurrent engineering)
- Maintains customer ideas and requirements, in the customer’s words, throughout the process

3. Benchmark on Customer Needs

#	NEED		Imp	ST Tritrack	Maniray 2	Rox Tahx Quadra	Rox Tahx Ti 21	Tonka Pro	Gunhill Head Shox
1	The suspension	reduces vibration to the hands.	3	•	••••	••	•••••	••	•••
2	The suspension	allows easy traversal of slow, difficult terrain.	2	••	••••	•••	•••••	•••	•••••
3	The suspension	enables high speed descents on bumpy trails.	5	•	•••••	••	•••••	••	•••
4	The suspension	allows sensitivity adjustment.	3	•	••••	••	•••••	••	•••
5	The suspension	preserves the steering characteristics of the bike.	4	••••	••	•	••	•••	•••••
6	The suspension	remains rigid during hard cornering.	4	•	•••	•	•••••	•	•••••
7	The suspension	is lightweight.	4	•	•••	•	•••	••••	•••••
8	The suspension	provides stiff mounting points for the brakes.	2	•	••••	•••	•••	••	•••••
9	The suspension	fits a wide variety of bikes, wheels, and tires.	5	••••	•••••	•••	•••••	•••	•
10	The suspension	is easy to install.	1	••••	•••••	••••	••••	•••••	•
11	The suspension	works with fenders.	1	•••	•	•	•	•	•••••
12	The suspension	instills pride.	5	•	••••	•••	•••••	•••	•••••
13	The suspension	is affordable for an amateur enthusiast.	5	•••••	•	•••	•	•••	••
14	The suspension	is not contaminated by water.	5	•	•••	••••	••••	••	•••••
15	The suspension	is not contaminated by grunge.	5	•	•••	•	••••	••	•••••
16	The suspension	can be easily accessed for maintenance.	3	••••	•••••	••••	••••	•••••	•
17	The suspension	allows easy replacement of worn parts.	1	••••	•••••	••••	••••	•••••	•
18	The suspension	can be maintained with readily available tools.	3	•••••	•••••	•••••	•••••	••	•
19	The suspension	lasts a long time.	5	•••••	•••••	•••••	•••	•••••	•
20	The suspension	is safe in a crash.	5	•••••	•••••	•••••	•••••	•••••	•••••

Benchmark on Metrics

Metric #	Need #s	Metric	Imp	Units	ST Tritrack	Maniray 2	Rox Tahx Quadra	Rox Tahx Ti 21	Tonka Pro	Gunhill Head Shox
1	1,3	Attenuation from dropout to handlebar at 10hz	3	dB	8	15	10	15	9	13
2	2,6	Spring pre-load	3	N	550	760	500	710	480	680
3	1,3	Maximum value from the Monster	5	g	3.6	3.2	3.7	3.3	3.7	3.4
4	1,3	Minimum descent time on test track	5	s	13	11.3	12.6	11.2	13.2	11
5	4	Damping coefficient adjustment range	3	N-s/m	0	0	0	200	0	0
6	5	Maximum travel (26in wheel)	3	mm	28	48	43	46	33	38
7	5	Rake offset	3	mm	41.5	39	38	38	43.2	39
8	6	Lateral stiffness at the tip	3	kN/m	59	110	85	85	65	130
9	7	Total mass	4	kg	1.409	1.385	1.409	1.364	1.222	1.1
10	8	Lateral stiffness at brake pivots	2	kN/m	295	550	425	425	325	650
11	9	Headset sizes	5	in	1.000 1.125	1.000 1.125 1.250	1.000 1.125	1.000 1.125 1.250	1.000 1.125	NA
12	9	Steertube length	5	mm	150 180 210 230 255	140 165 190 215	150 170 190 210	150 170 190 210 230	150 190 210 220	NA
13	9	Wheel sizes	5	list	26in	26in	26in	26in 700C	26in	26in
14	9	Maximum tire width	5	in	1.5	1.75	1.5	1.75	1.5	1.5
15	10	Time to assemble to frame	1	s	35	35	45	45	35	85
16	11	Fender compatibility	1	list	Zefal	none	none	none	none	all
17	12	Instills pride	5	subj	1	4	3	5	3	5
18	13	Unit manufacturing cost	5	US$	65	105	85	115	80	100
19	14	Time in spray chamber w/o water entry	5	s	1300	2900	>3600	>3600	2300	>3600
20	15	Cycles in mud chamber w/o contamination	5	k-cycles	15	19	15	25	18	35
21	16,17	Time to disassemble/assemble for maintenance	3	s	160	245	215	245	200	425
22	17,18	Special tools required for maintenance	3	list	hex	hex	hex	hex	long hex	hex, pin wrnch
23	19	UV test duration to degrade rubber parts	5	hours	400+	250	400+	400+	400+	250
24	19	Monster cycles to failure	5	cycles	500k+	500k+	500k+	480k	500k+	330k
25	20	Japan Industrial Standards test	5	binary	pass	pass	pass	pass	pass	pass
26	20	Bending strength (frontal loading)	5	MN	55	89	75	75	62	102

4. Set Ideal and Acceptable Values

	Metric	Units	Marginal Value	Ideal Value
1	Attenuation from dropout to handlebar at 10hz	dB	>10	>15
2	Spring pre-load	N	480 - 800	650 - 700
3	Maximum value from the Monster	g	<3.5	<3.2
4	Minimum descent time on test track	s	<13.0	<11.0
5	Damping coefficient adjustment range	N-s/m	0	>200
6	Maximum travel (26in wheel)	mm	33 - 50	45
7	Rake offset	mm	37 - 45	38
8	Lateral stiffness at the tip	kN/m	>65	>130
9	Total mass	kg	<1.4	<1.1
10	Lateral stiffness at brake pivots	kN/m	>325	>650
11	Headset sizes	in	1.000 1.125	1.000 1.125 1.250
12	Steertube length	mm	150 170 190 210	150 170 190 210 230
13	Wheel sizes	list	26in	26in 700c
14	Maximum tire width	in	>1.5	>1.75
15	Time to assemble to frame	s	<60	<35
16	Fender compatibility	list	none	all
17	Instills pride	subj	>3	>5
18	Unit manufacturing cost	US$	<85	<65
19	Time in spray chamber w/o water entry	s	>2300	>3600
20	Cycles in mud chamber w/o contamination	k-cycles	>15	>35
21	Time to disassemble/assemble for maintenance	s	<300	<160
22	Special tools required for maintenance	list	hex	hex
23	UV test duration to degrade rubber parts	hours	>250	>450
24	Monster cycles to failure	cycles	>300k	>500k
25	Japan Industrial Standards test	binary	pass	pass
26	Bending strength (frontal loading)	MN	>70	>100

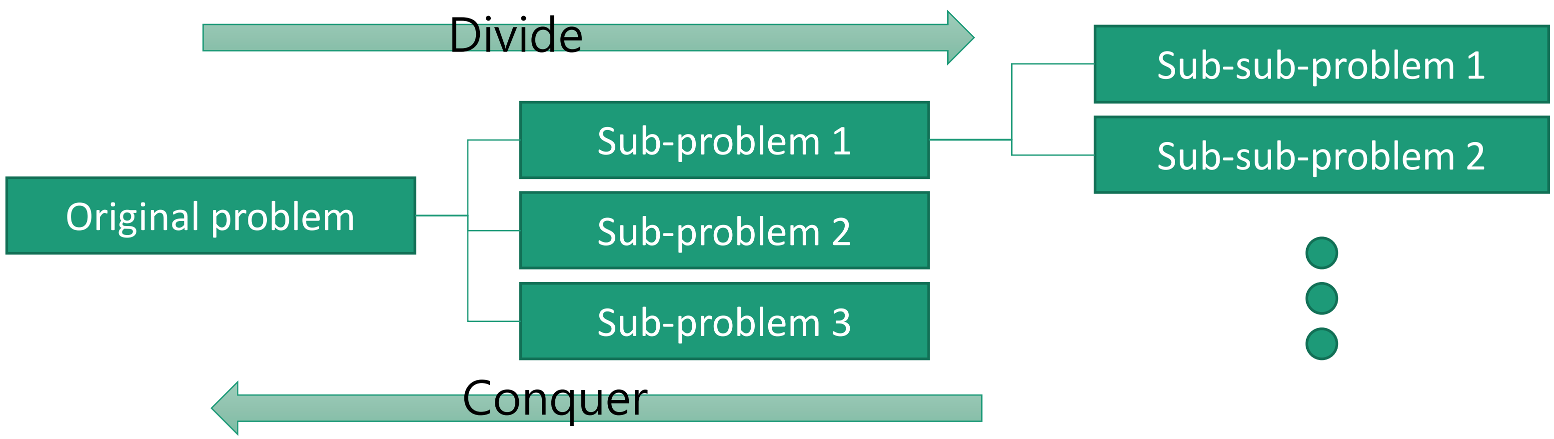
Homework Assignments

- Read chapter 6 in PD&D book
- Complete Individual Assignment 2

Student Activities

Divide and Conquer

- Originally, the term described a specific war strategy
- To break down a complicated problem into simple sub-problems whose solutions can be tackled directly. Solution of the original problem is obtained by the combination of the solutions of sub-problems.
- Well fitted to the nature of human thinking
- Top-Down Approach (cf. Bottom-Up Approach)



Divide and Conquer

- Divide based on various levels (from conceptual/abstract to product/detail)
- Each level should have proper level-of-abstraction(LOA)
- For each level, the problem should be divided into sub-problems of similar size/complexity

Divide and Conquer

- Divide more
  - If knowledge-requirements exceed prior expectations
  - If the final solution is more complicated and requires more novelty
- Stop Divide
  - If you can propose solutions for the divided parts of the problem
  - If elements of a problem is closely related each other (=cohesive) and divided parts require many interactions among themselves

Divide and <u>Conquer</u>

- Solve the sub-problems, after divide the original problem into sub-problems whose solutions are feasible.
- Combine the solutions of the sub-problems to solve the original problem
- In most cases, the solution of the sub-problem can be obtained easier than the solution of the whole problem

Divide and Conquer

- Especially good for open problem solving
- Not confined to engineering; widely used in various fields
- Limitation
  - If the sub-problems are strongly related each other so that each sub-problem cannot be solved independently, divide and conquer may not be a good approach
  - Examples: Size reduction of cell phones, Car fuel-efficiency (Miles per gallon) enhancement, etc.

Divide and Conquer Problem

- Assume that we are constructing a housing facility on the moon to accommodate at least 10 people. Identify all works required for the construction.

  - Breakdown the whole works into several groups
  - For each group, define detailed activities
  - If required, each group can be broken down into smaller sub-groups.

Divide and Conquer Problem

- Estimate the whole cost you spend during your university study.

  - Breakdown the whole cost into several categories
  - For each category, estimate the cost.
  - If required, each category can be broken down into sub-categories.

Interpret Customer Needs and "Guess" Importance

Customer statements about a student backpack:

- See how the leather on the bottom of the bag is all scratched; it's ugly.
- When I'm standing in line at the cashier trying to find my checkbook while balancing my bag on my knee, I feel like a stork.
- This bag is my life; if I lose it I'm in big trouble.
- There's nothing worse than a banana that's been squished by the edge of a textbook.
- I never use both straps on my knapsack; I just sling it over one shoulder.

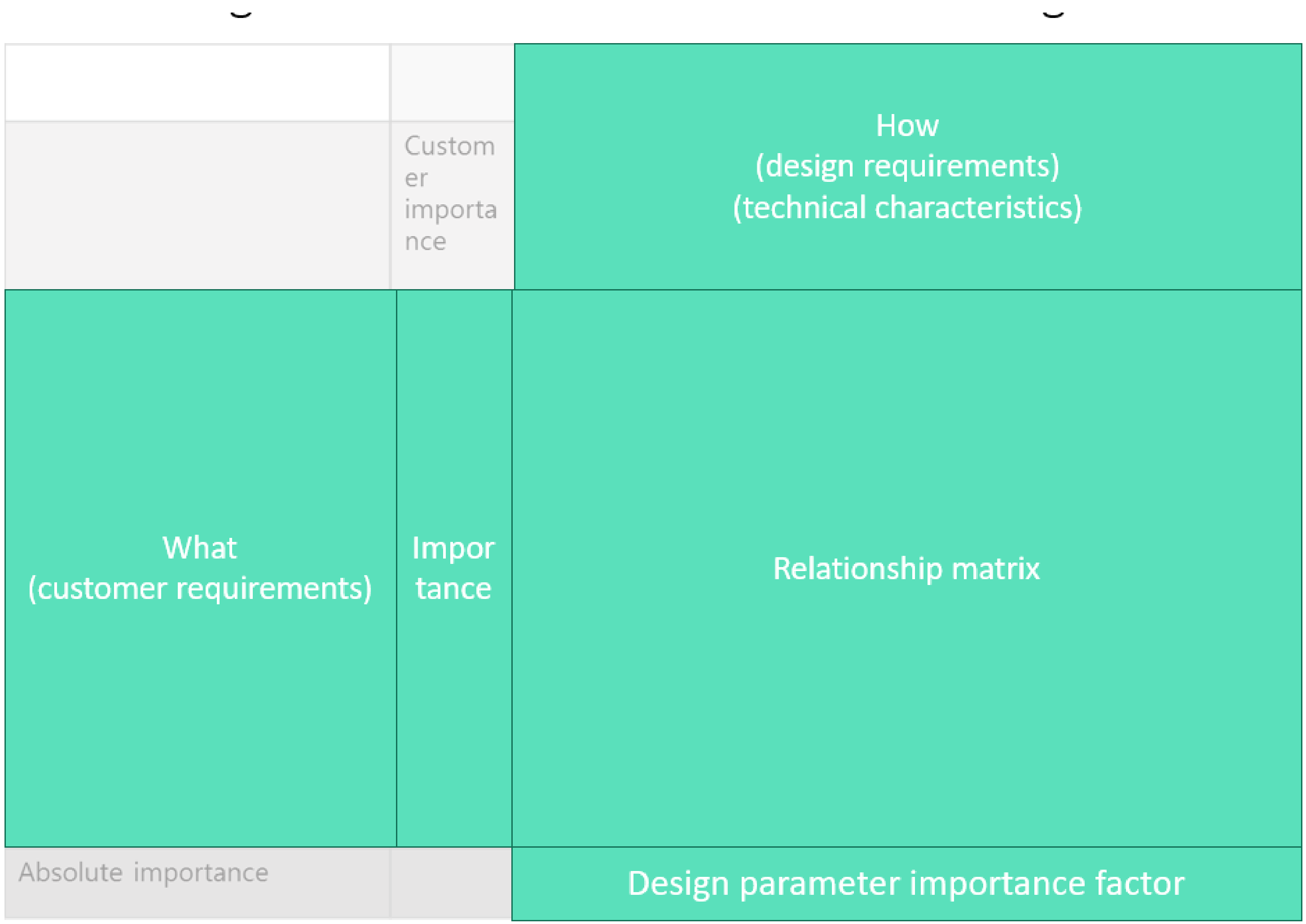
Establish Quality Metrics and Units of Measurement for Backpack

- Function/Performance
- Product Cost
- Lead time (Time-to-market)
- Quantity
- Environmental effect
- Safety
- Quality
- Energy efficiency
- Reliability
- Maintenance
- Mechanical weight
- Size
- Space constraints
- Aesthetics
- Packaging and delivery
- Human resource
- Product life time
- Operation noise
- Operating guide
- Human factors
- Health
- Government regulations
- Operation cost

Benchmarking with Competitor Products is Missed Here

You can do this procedure as a home exercise

Create QFD for Backpack



Set Ideal and Acceptable Values

- Identify 3 key metrics that impact the customer requirements the most.
- Establish ideal and acceptable values for your backpack metrics.
  - Note, that on these 3 key metrics you should be better than your competitors
  - Other metrics should not be below acceptable values.
- Compile full technical specifications of your new product.

Kidney Exchanges

Engineering Idea #3

Mechanisms without Money

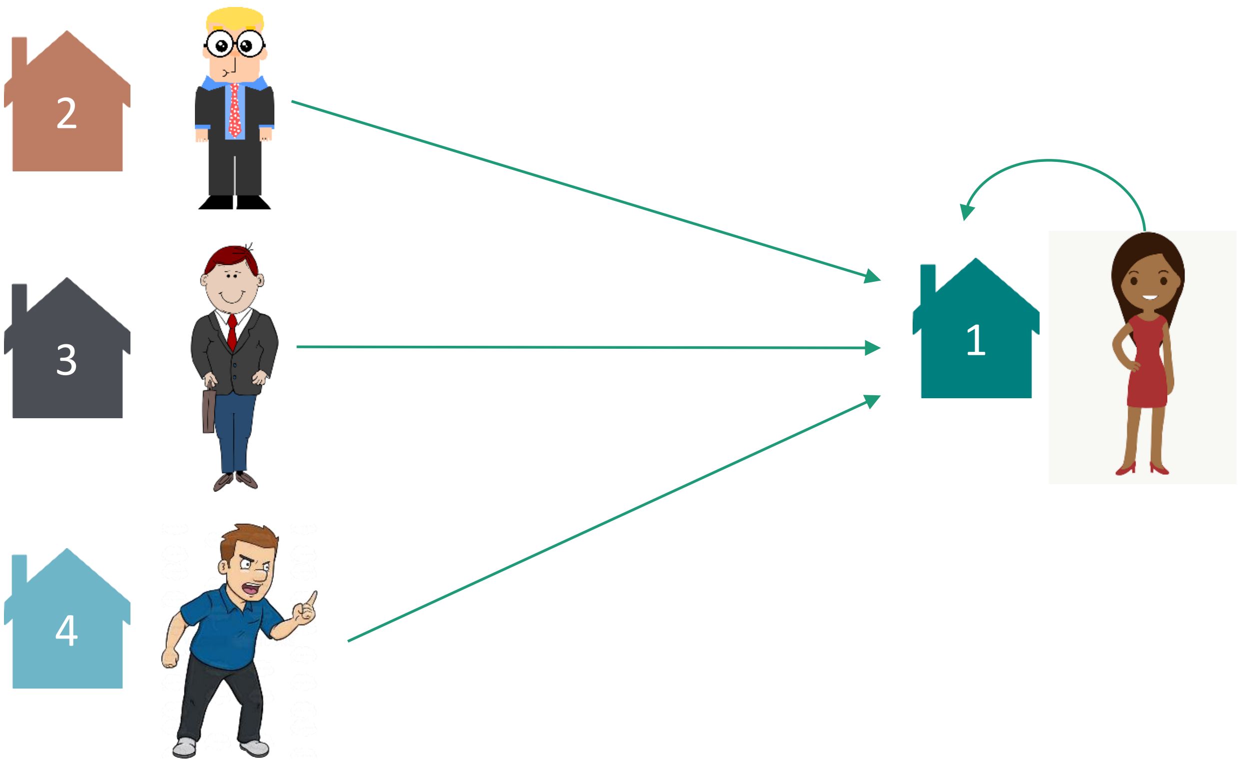
- Economics is a discipline which studies the allocation of limited resources for the satisfaction of human needs.
- Usually, this process occurs with the use of money as an intermediary.
- However, use of money can be infeasible, forbidden or unethical in some situations
- Examples: marriage problem, organ donors, university applications, house allocation

House Allocation Problem

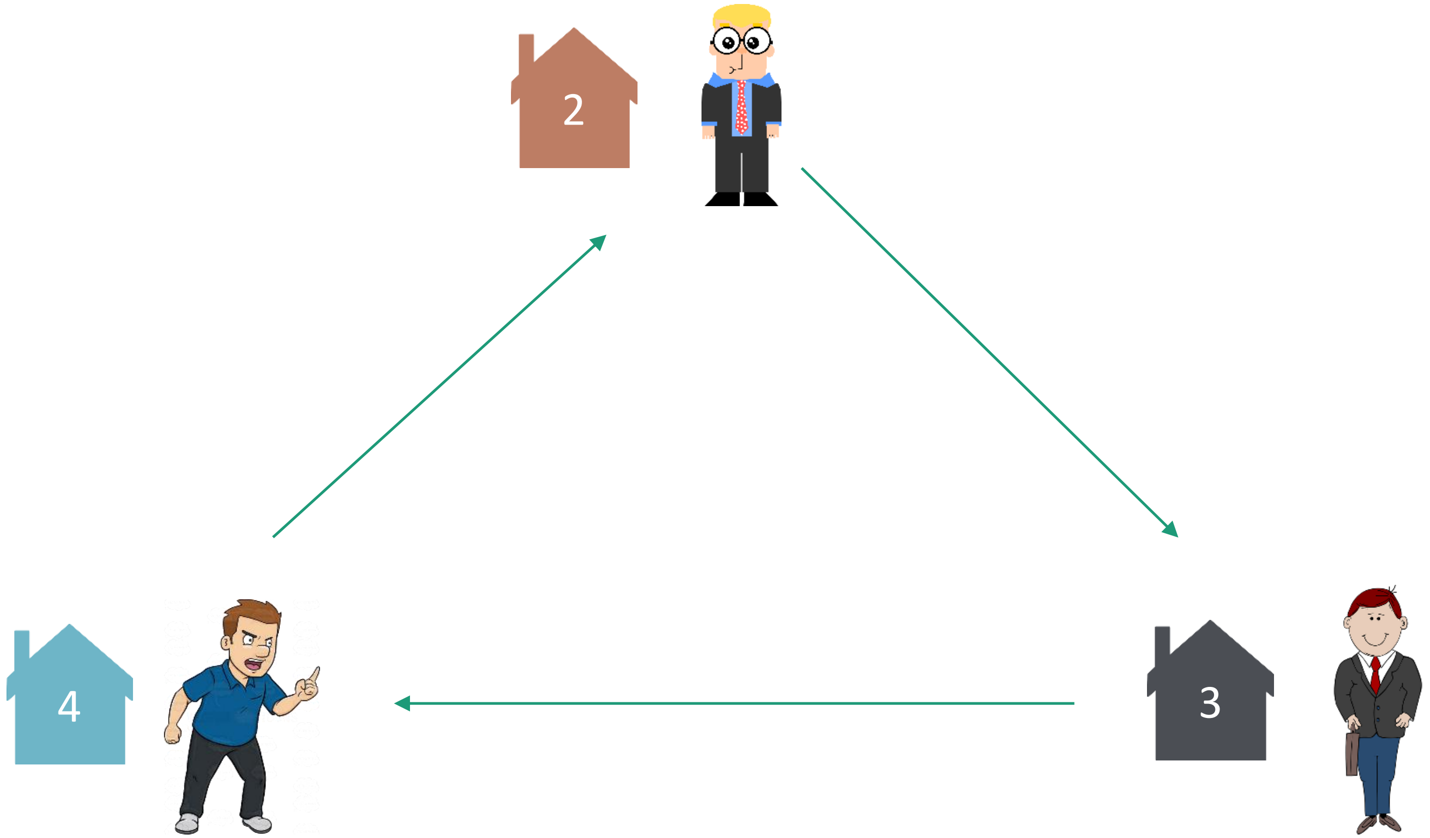
Top Trading Cycle (TTC) Algorithm

1. Initialize all players with corresponding houses.
2. If players remaining, let each remaining player point to her favourite (top) remaining house. It is a Directed Graph where each vertex has out-degree 1. Otherwise, exit.
3. Find cycles in the Directed Graph.
4. Reallocate as suggested by the directed cycles in the graph (including self-loops).
5. Remove reallocated players and houses from the list.
6. Go to step 2.

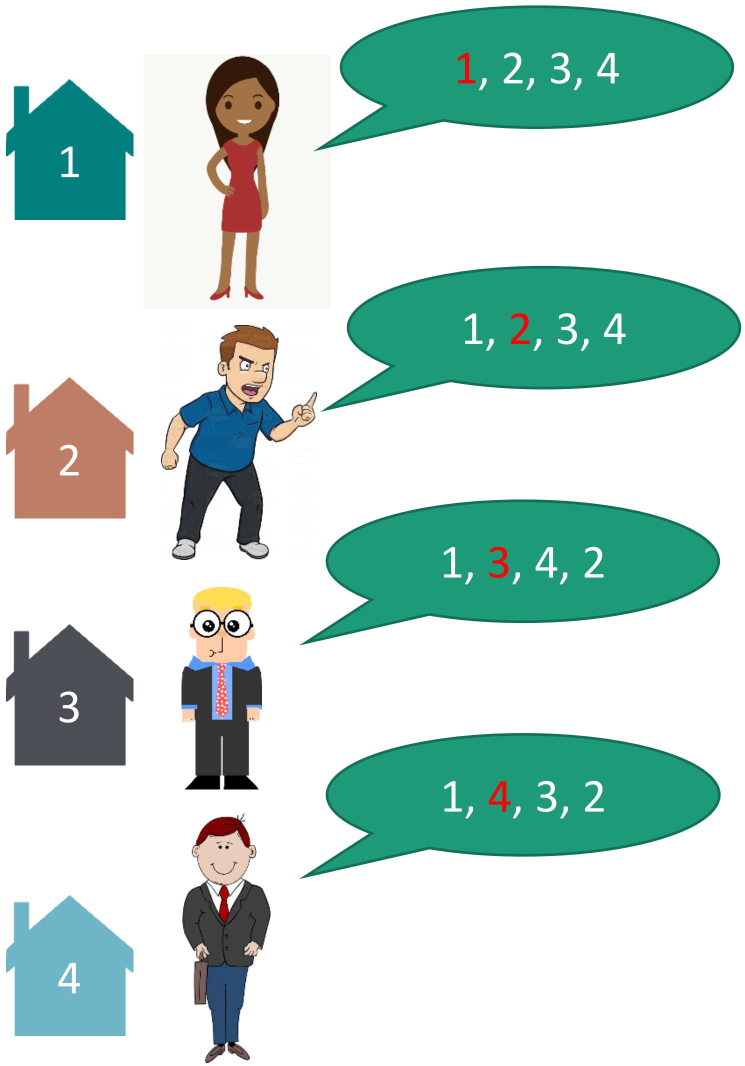
First Iteration



Second Iteration



Final Allocation



Properties of TTC Algorithm

- Termination
  - The algorithm is guaranteed to terminate, because there is always a cycle in the graph and it reduces # players.
  - The reason for the existence of a cycle is the fact that each vertex has exactly one outgoing edge, and therefore we can start at any vertex and follow outgoing edges as long as we wish.
  - At some point, a vertex must repeat, and a cycle is found.
- Weakly improved allocation
  - At the end, each agent has a house that she likes at least as much as her initial house

Properties of TTC Algorithm

- Incentive compatibility (aka Truthfulness is the best strategy)
  - Any individual player cannot change the outcome of the allocation by picking different houses (other than what she wants privately)
- Optimality
  - After allocation there is no group of agents who would want to reallocate.

Kidney Exchange Problem

- Kidney transplantation – living donor gives his/her kidney to a recipient.
- Usually, it is done between close relatives.
- Problem: their organs are not suitable to each other
- But kidney of a willing donor can be suitable to other patient.
- It can be solved similar to house allocation where donor-recipient pairs (as vertex) are connected by their compatibility to other donor-recipient pairs.